The documentation and process conversion measures necessary to comply with this amendment shall be completed by 31 October 2000.

INCH POUND

MIL-PRF-19500/368D
AMENDMENT 1
31 July 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES: 2N3439, 2N3439L, 2N3440, AND 2N3440L
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This amendment forms a part of MIL-PRF-19500/368D, dated 30 June 1999 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, Delete in it's entirety and substitute,

1.2. Physical dimensions. See figure 1 (similar to TO-5 and TO-39), figure 2 (JANHCA and JANKCA (A versions)), and figure 3 (JANHCB and JANKCB (B versions)).

1.3 Maximum ratings. Delete and replace with;

"1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C."

Types	P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_{STG} and T_{OP}	$R_{\theta JA}$
	W	W	V dc	V dc	V dc	A dc	°C	°C/W
2N3439	0.8	5.0	450	7	350	1.0	-65 to +200	325
2N3440	0.8	5.0	300	7	250	1.0	-65 to +200	325

1/ Derate linearly 5.7 mW/°C for T_A > +60°C.
2/ Derate linearly 28.6 mW/°C for T_C > +25°C."

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Drawing 2 change the title to read; "FIGURE 2. Physical dimensions JANHCA and JANKCA (die) A version."

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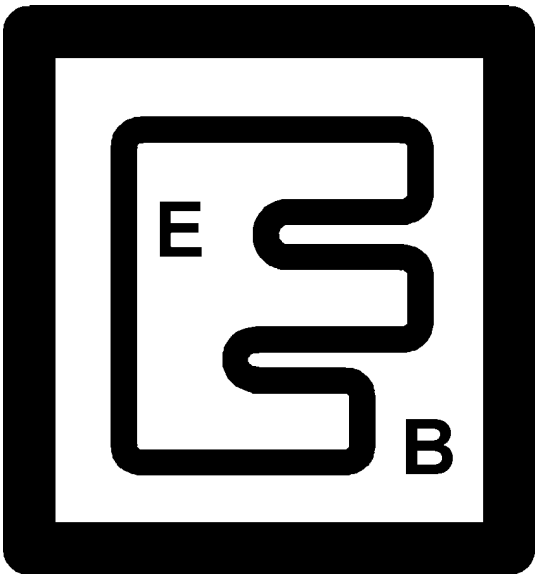
Add figure 3.



1. Chip size.....0.040 x 0.040 inches ±0.002 inches
2. Chip thickness.....0.010 ± 0.0015inches nominal
3. Top metal.....Aluminum 30,000Å minimum, 33,000Å nominal
4. Back metal.....A. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅminimum,15kÅ/ 5kÅ/10kÅ/10kÅ nominal.
   B. Gold 2.500Å minimum, 3000Å nominal
5. Backside.....Collector
6. Bonding pad.....B = 0.005 x 0.008 inches, E = 0.010 x 0.007 inches

FIGURE 3. Physical dimensions JANHCB and JANKCB (die) B versions.

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3.4. Interface requirements and physical dimensions Delete and replace with;

"3.4. Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (similar to TO-5 and TO-39), 2 and 3 (JANHC and JANKC) herein."

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4.3.2. Power burn-in conditions. Delete 4.3.2 completely and replace with;

"4.3.2. Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 Vdc. Power shall be applied to achieve T_j =135°C minimum and a minimum P_D = 75% of P_T maximum rated as defined in 1.3 herein.

4.3.3. Thermal impedance ($Z_{\theta JX}$ measurements), c. t_H heating time, Delete the "100 ms" and replace with "10 ms".

4.4.2.1. Group B inspection, table VIa (JANS) of MIL-PRF-19500. Delete and replace with;

"4.4.2.1. Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc.
B5	1027	V_{CB} = 10 V dc; T_A = +125°C ±25°C for 96 hours with P_T adjusted according to the chosen T_A to give T_j = +275°C minimum. Optionally the test may be performed for a minimum of 216 hours with P_T adjusted to achieve a T_j = +225°C; sample size = 45, c = 0. In this case the ambient temperature shall be adjusted such that a minimum 75% of maximum rated P_T (see1.3) is applied to the device under test. (Note: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

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4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV), Step 1, Condition, Delete conditions and replace with;

"Steady-state life: Test condition B, 340 hours, V_{CB} = 10 -30 V dc, Power shall be applied to achieve T_J = 150°C minimum and a minimum of P_D = 75% of maximum rated P_T as defined in 1.3 herein. n = 45 devices, c = 0"

4.4.3.1. Group C inspection, table VII (JANS) of MIL-PRF-19500, Subgroup C6. Delete and replace with;

"1026 1,000 hours at V_{CB} 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75% of maximum rated P_T as defined in 1.3 herein."

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4.5.3 Delta requirements. Step 2, conditions, delete " I_C = 2.0 mA" and substitute "I_C = 20 mA".

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2204)

Review activities:
Army - MI, SM
Navy - AS, CG, MC
Air Force - 19, 99

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